

Ip address and subnet

1. List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

IP Class	Address Range	Purpose	Real-World Example
Class A	1.0.0.0 – 126.255.255.255	Large networks	A large organization's internal network using a private Class A range such as 10.0.0.0/8
Class B	128.0.0.0 – 191.255.255.255	Medium-sized networks	A university or college network using a Class B-style network
Class C	192.0.0.0 – 223.255.255.255	Small networks	A home/office LAN using 192.168.1.0/24
Class D	224.0.0.0 – 239.255.255.255	Multicast communication	224.0.0.1 , used for multicast communication between network devices
Class E	240.0.0.0 – 255.255.255.255	Experimental/reserved	Used for experimental or research purposes , not normally assigned to regular devices

2. Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses

- IP Address: 192.168.10.45
- Subnet Mask: 255.255.255.0 (/24)

Step 1: Network Address Set all host bits to 0. Network Address = 192.168.10.0

Step 2: Broadcast Address Set all host bits to 1. Broadcast Address = 192.168.10.255

Step 3: Valid Host Range Exclude the network and broadcast addresses. Valid Hosts = 192.168.10.1 to 192.168.10.254

Step 4: Total Valid Hosts $2^8 - 2 = 254$ hosts

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3. Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and

one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

Given: 10.0.0.0/24 and **4 subnets required.**

Step 1: Borrow 2 host bits because $2^2 = 4$ subnets.

$$/24 + 2 = /26$$

Step 2: New subnet mask:

$$/26 = 255.255.255.192$$

Step 3: Subnet block size:

$$256 - 192 = 64$$

So, the networks increase by **64**:

Group	Network Address	Subnet Mask
Delivery Partners	10.0.0.0	255.255.255.192 (/26)
Restaurants	10.0.0.64	255.255.255.192 (/26)
Customers	10.0.0.128	255.255.255.192 (/26)
Admins	10.0.0.192	255.255.255.192 (/26)

Each subnet: 64 addresses → **62 usable host addresses.**

4. You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet.

Given: 172.16.0.0/16

Required: At least **20 subnets**

Step 1: Borrow bits

We need:

- $2^4 = 16$ ✗ not enough
- $2^5 = 32$ ✓ enough

So, borrow **5 bits.**

Step 2: New subnet mask

$$/16 + 5 = /21$$

$$/21 = 255.255.248.0$$

Step 3: Hosts per subnet

$$\text{Remaining host bits} = 32 - 21 = 11$$

Usable hosts:

$$2^{11} - 2 = 2046$$

5. A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

The IP address **192.168.1.300** is invalid because each part of an IPv4 address must be between **0 and 255**. The value **300** exceeds the maximum allowed value.

For the **192.168.1.0/24** subnet, a valid host IP would be between **192.168.1.1** and **192.168.1.254**, for example **192.168.1.100**.